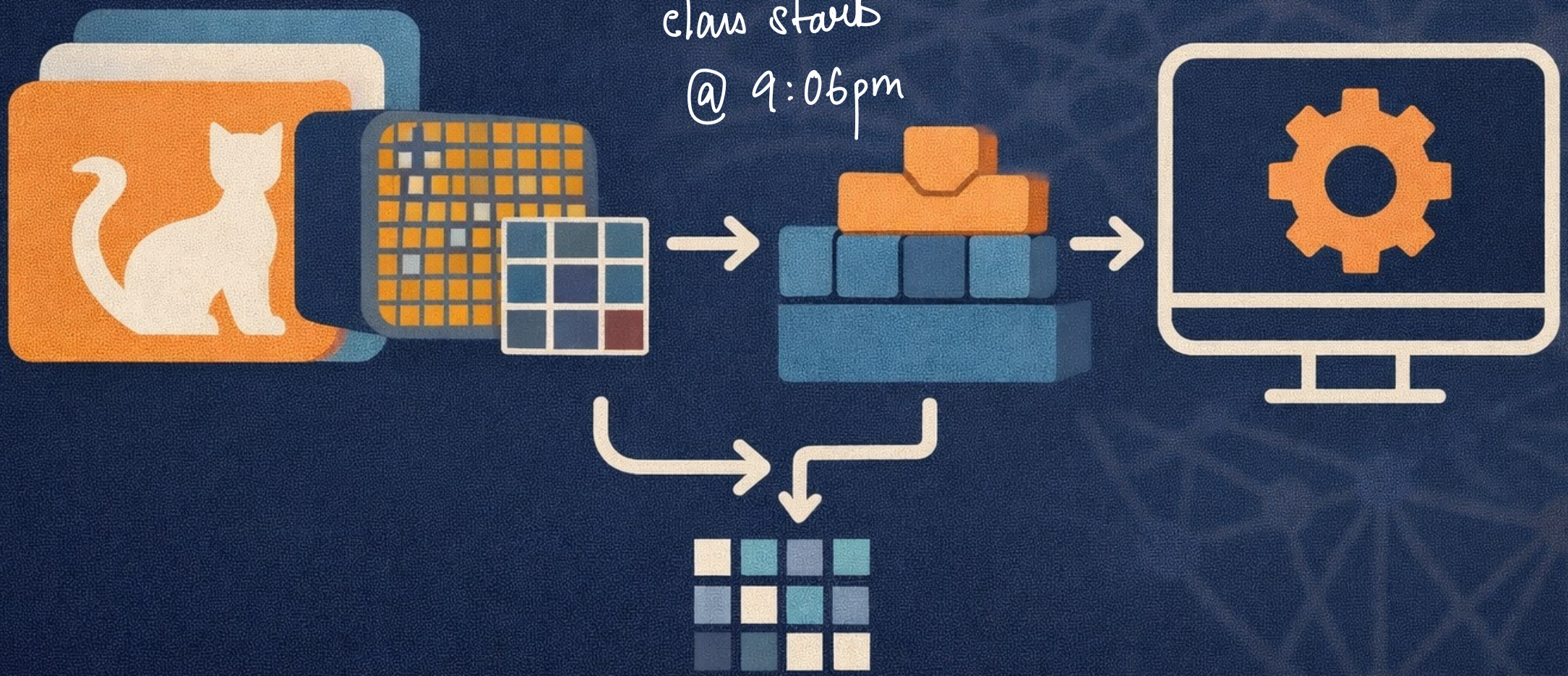
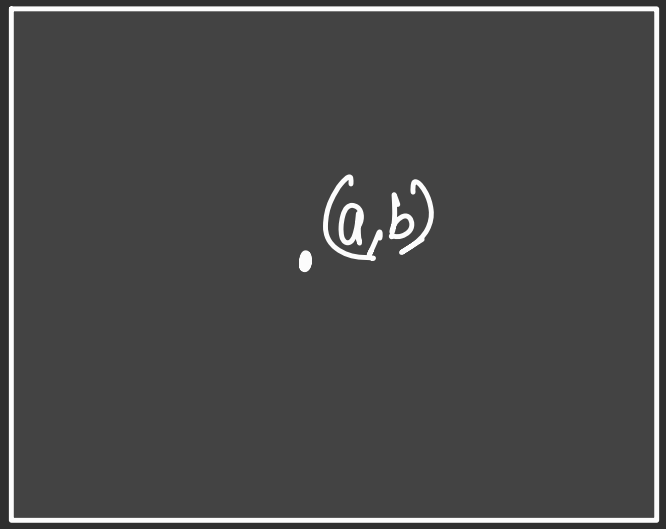
CONVOLUTIONAL NEURAL NETWORK



Lecture : Object detection 2

#Agenda

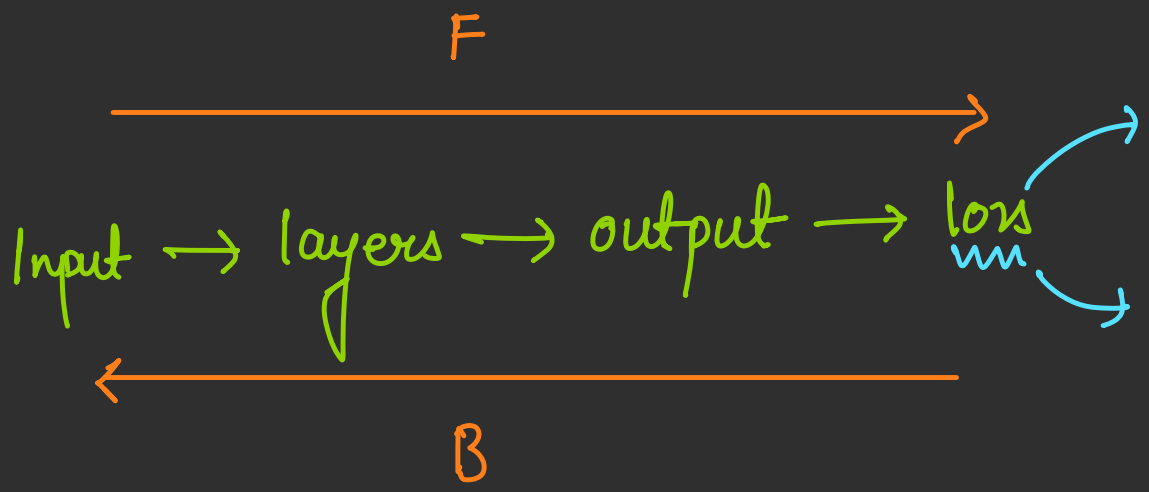
1) Object Localization
2) Object detection
3) RCNN & Fast RCNN

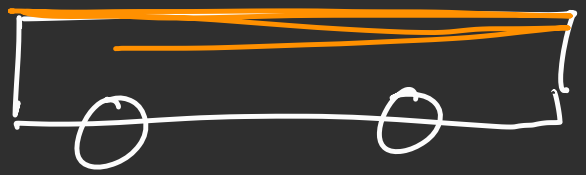


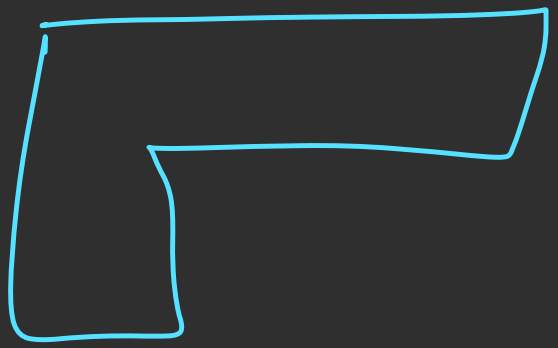
a b c d

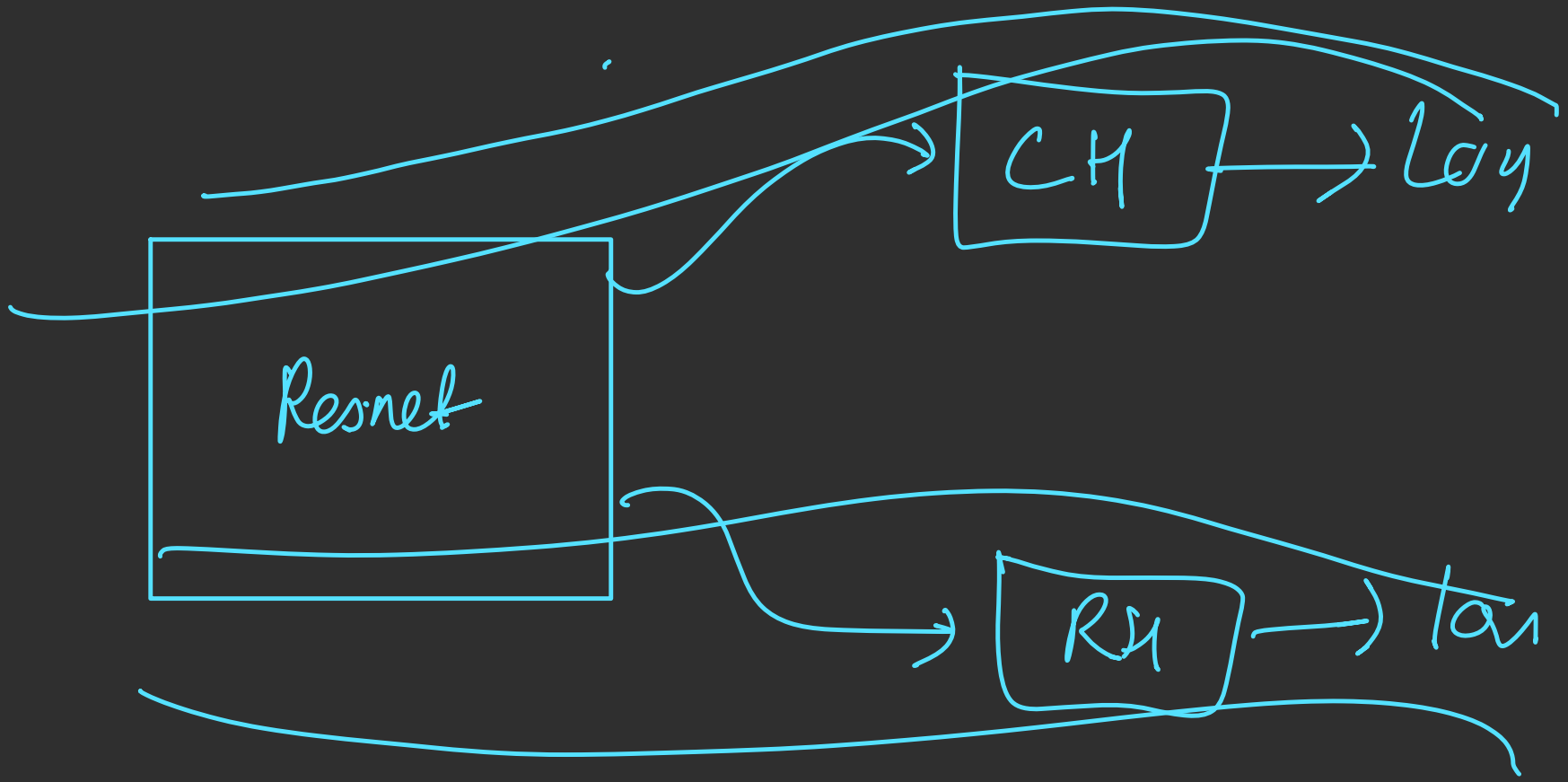
c → width

d → height

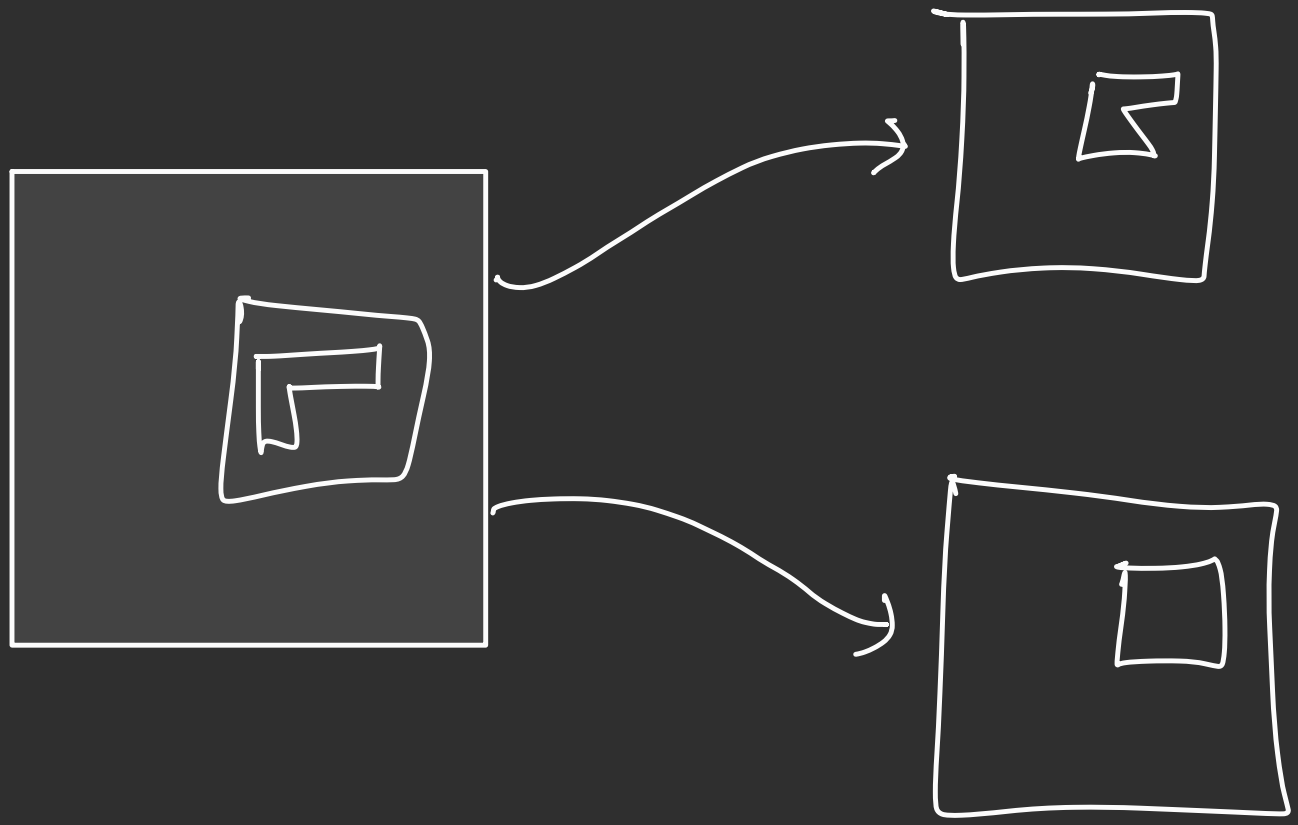

F
Input
layers
output
loss
B

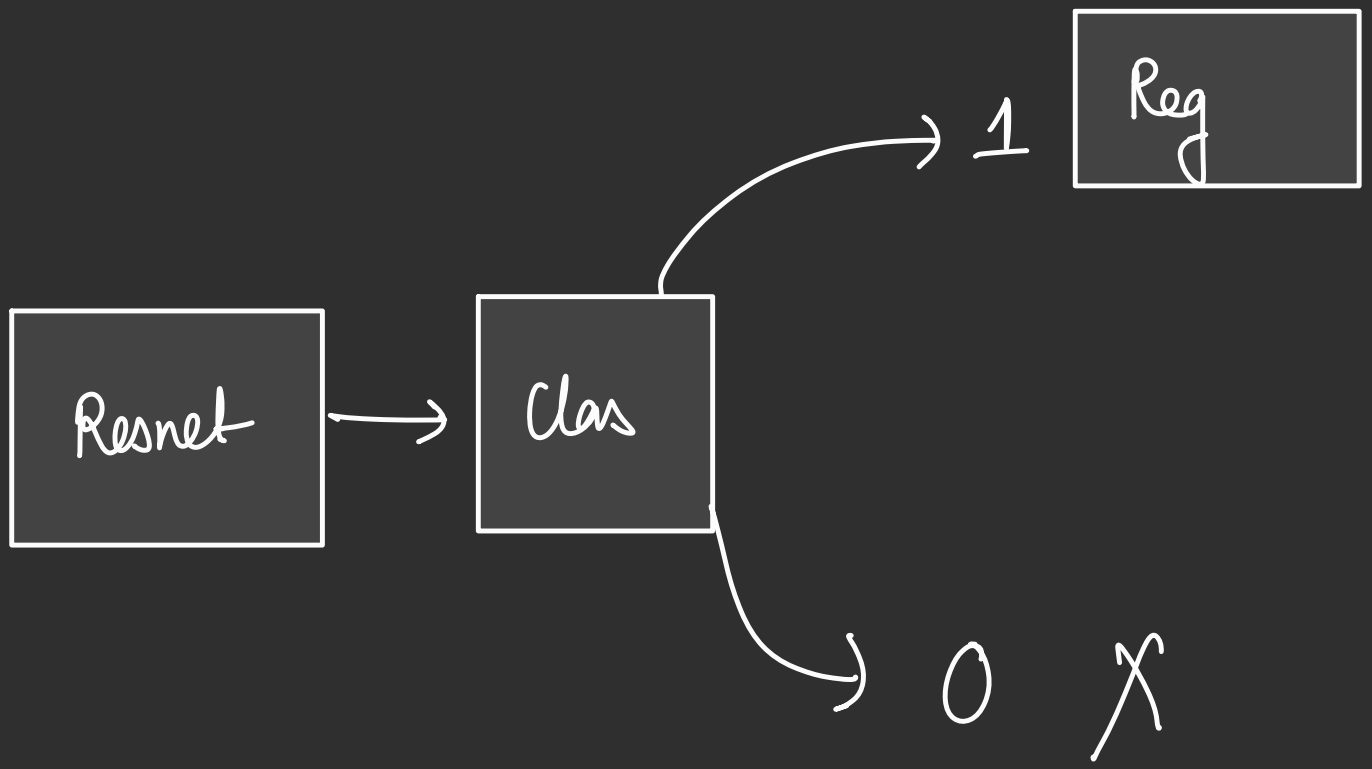






MSE + Binary CE



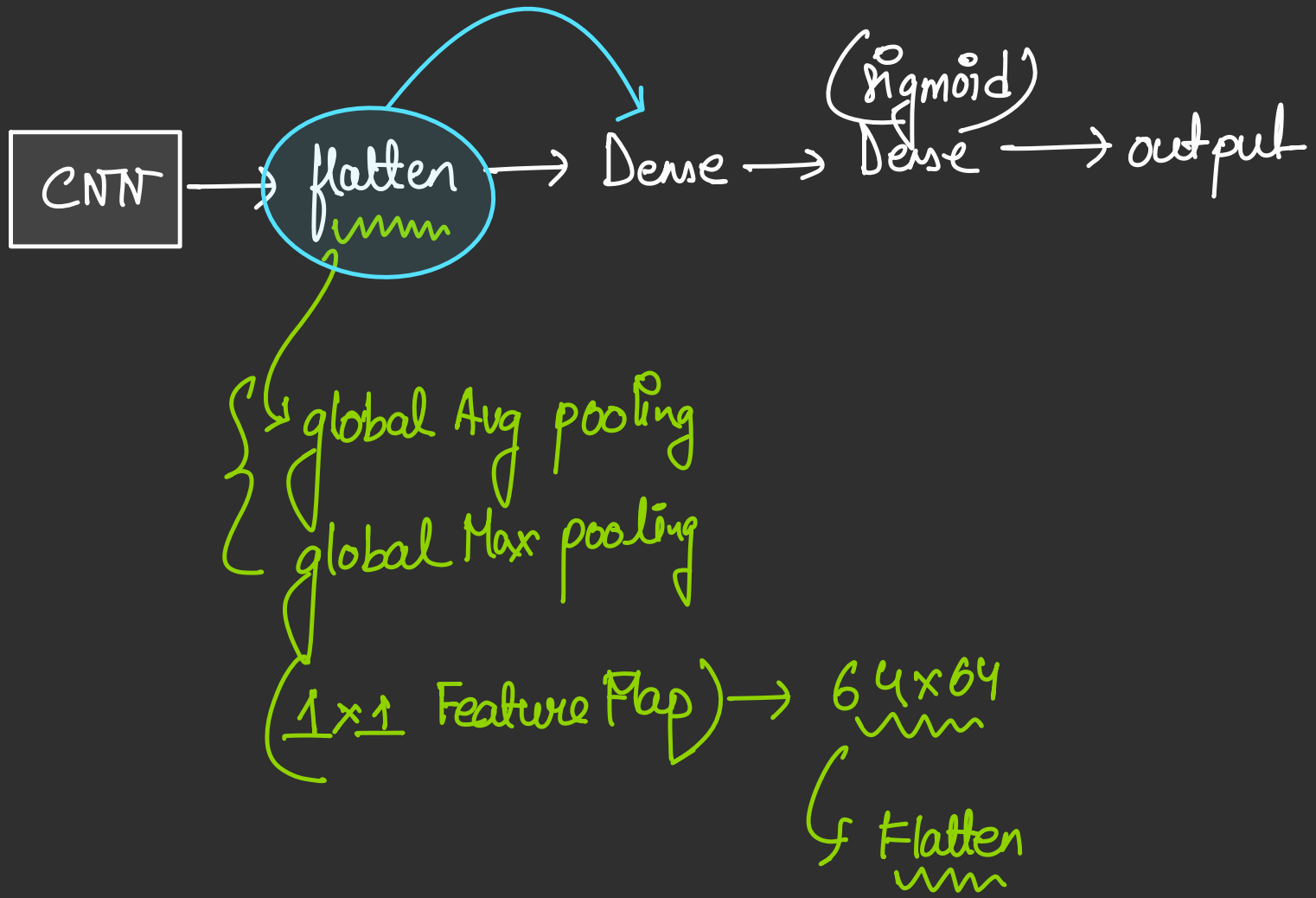

Resnet
Clas
1
Reg
0
X

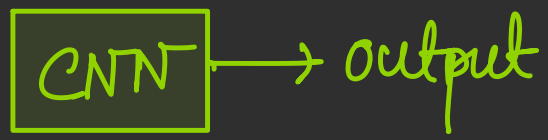
3 Ways

1) Random problem → apply the coding (Hands on)

2) Take a specific problem statement (IC, OD) → Transfer learning
   [Flow: preprocessing steps, evaluation of that model] → Output

3) Do things from scratch


CNN
flatten
Dense
Dense
sigmoid
output
global Avg pooling
global Max pooling
1x1 Feature Map
64x64
Flatten



$[128 \times 128 \times 3] \rightarrow$ 1×1 Conv $\rightarrow (128 \times 128)$